

Haiyang Zhang

Senior Full-Stack Engineer

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PROFESSIONAL SUMMARY

- Senior Full-Stack Engineer and founding engineer at Lavita AI, with 8 years of experience building web, mobile, and AI applications.
- Built core features for a HIPAA-compliant healthcare app with 100,000+ downloads and an enterprise workflow platform delivered under a \$5.9M contract.
- Led a six-person team and improved delivery through platform refactoring, shared development standards, and code reviews.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL, HTML5, CSS3
- **Frontend and Mobile:** React (React.js), Next.js, React Native, Expo, Vue.js, Nuxt.js, Redux, Zustand, Vuex, Pinia, Tailwind CSS, Sass/SCSS
- **Backend and Data:** Node.js, Express.js, FastAPI, PostgreSQL, RESTful APIs, GraphQL
- **AI:** Retrieval-Augmented Generation (RAG)
- **Real-Time and Streaming:** Server-Sent Events (SSE), WebSocket
- **Healthcare and Interoperability:** HIPAA, Protected Health Information (PHI), Electronic Health Records (EHR), HL7, FHIR, SMART on FHIR, Epic, Cerner, OAuth 2.0
- **Cloud and DevOps:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Docker, Nginx, CI/CD
- **Testing:** Jest, Cypress, Playwright

PROFESSIONAL EXPERIENCE

Lavita AI

Senior Full-Stack Engineer | Sep 2018 - Present

Healthcare Mobile App, AI Services, and EHR Integration

- Delivered AI chat, health-data upload, and account workflows for a React Native healthcare app with 100,000+ downloads across iOS and Android.
- Built REST APIs for AI chat, health-data processing, and account management using Node.js and Python.
- Built a retrieval-augmented generation (RAG) pipeline to supply healthcare context for large language model (LLM) responses.
- Reduced AI chat time to first token (TTFT) by 40% using SSE streaming, verified through frontend and backend measurements.
- Implemented AI output validation and bounded retries with corrective feedback for responses that failed quality checks.
- Integrated electronic health records (EHR) from Epic and Cerner using OAuth 2.0 and SMART on FHIR, with HIPAA-compliant PHI handling and HL7/FHIR transformations that reduced medical-record payload size by 70%.
- Expanded automated test coverage for critical app workflows to 65%+, reducing release-blocking regression frequency by 35%.

Enterprise Workflow Platform Development

- Owned compute node management and data collaboration frontends for a B2B workflow platform delivered under a \$5.9M contract.
- Built interfaces for node connectivity, resource and container monitoring, access control, and logs.

- Implemented data-source registration and preprocessing workflows for structured and unstructured data.
- Built workflow orchestration interfaces for visual editing, execution tracking, and model and result publishing.
- Automated regression tests for 20+ critical workflows, reducing manual regression testing time by 40%.

Enterprise Workflow Platform Refactoring

- Led refactoring of two subsystem families into a reusable foundation for ongoing maintenance and customer-specific delivery.
- Consolidated 10+ recurring patterns into reusable modules for role-based access control (RBAC), forms, approvals, and audit logs.
- Standardized six reusable workflow templates for project setup, task management, execution tracking, and result publishing.
- Reduced duplicated implementation by 30-40% by consolidating shared components, data models, and workflow logic.
- Standardized component contracts, build practices, onboarding, and code reviews for a six-person team.

EDUCATION

Syracuse University

M.S.E. in Computer and Information Science | 2018

Beijing University of Technology

B.S.E. in Internet of Things | 2016